Efficient Latent Bridge Matching for Domain-Level Artistic Transfer

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Abstract

Efficient artistic style transfer remains difficult because style strength, content preservation, and computational cost often conflict. We study domain-level multi-style artistic transfer and propose a latent bridge-inspired framework that transports compact VAE latents with a style-conditioned velocity field. The method combines terminal sliced-Wasserstein distribution matching, which drives the endpoint toward the target artistic domain, with kinetic regularization, which penalizes excessive latent displacement and preserves content structure. A learnable style spatial prior and semantic cross-attention provide efficient domain-level style conditioning without per-image optimization or diffusion sampling. Under a strict 750-output protocol over five domains, the proposed model reaches a competitive raw CLIP-style score of 0.7161, improves LPIPS-content over SaMST, and substantially improves artifact-sensitive perceptual diagnostics including MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, and shallow Gram statistics. Additional experiments complete strict-750 core-metric comparisons to StyTr2, CAST, AesFA, and AesPA-Net, and add step-count and $\lambda_{\mathrm{kin}}/\lambda_{\mathrm{swd}}$ sensitivity analyses. These experiments show that the method operates in a broad trade-off region rather than a brittle protocol-specific optimum. Four Euler steps already approach the 12-step default, while larger kinetic weight and lighter terminal pressure improve content preservation in a predictable way. The results support a conservative claim: latent bridge matching provides a clean, efficient, and reproducible style-content trade-off for domain-level artistic transfer, especially relative to heavier diffusion-style or optimization-heavy alternatives.

Introduction

Artistic style transfer has long served as a testbed for separating image content from visual appearance. Classical neural style transfer showed that convolutional features and Gram statistics can synthesize compelling artistic textures (Gatys, Ecker, and Bethge 2016), while feed-forward arbitrary style transfer made real-time deployment practical through normalization-based style modulation (Huang and Belongie 2017). More recent attention, transformer, training-free, and diffusion-based systems have improved flexibility and style fidelity (Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Huang et al. 2024; Gim, Yoo, and Uh 2024; Chung, Hyun, and Heo 2024). However, practical multi-style artistic transfer still exposes a persistent three-way tension: strong target-style expression, faithful content preservation, and low computational cost.

This tension is especially visible in domain-level multi-style transfer. Efficient multi-style systems can support many styles and fast inference, but the output may become washed, over-smoothed, or contaminated by structured grain artifacts. Diffusion-based or training-free methods can produce strong style semantics, but may be expensive or suffer semantic drift under content-preserving evaluation. Under our strict 750-output protocol, StyleID attains the highest raw CLIP-style score but exhibits severe content collapse, while SaMST achieves strong raw style and content scores yet produces visible muddy and grain-like texture artifacts. These observations suggest that standard style/content metrics alone are insufficient. A competitive system must also be evaluated for perceptual quality and artifact-sensitive texture realism.

We propose an efficient latent-space multi-style artistic transfer framework that models stylization as a style-conditioned latent probability flow. Instead of directly predicting RGB images, the method transports a compact VAE latent through a neural velocity field. A terminal sliced-Wasserstein distance (SWD) term aligns generated endpoints with the target artistic domain, while kinetic regularization penalizes excessive latent displacement and preserves content. A learnable style spatial prior and semantic cross-attention provide a lightweight approximation to local style transport without per-image optimization or diffusion sampling.

We call the resulting system *Latent Bridge Matching* (LBM). The name emphasizes that the method is not merely a feed-forward style injector: it matches the terminal artistic distribution while regularizing the latent path used to reach that distribution.

Our contributions are:

- We formulate domain-level multi-style artistic transfer as a bridge-inspired latent transport problem with separate terminal style matching and path-energy content preservation.
- We introduce an efficient style spatial prior and semantic cross-attention architecture that supports fast style-conditioned latent velocity prediction.

- We provide a strict 750-output evaluation protocol with complete core-metric comparisons to AdaIN, S2WAT, SaMST, StyleID, StyTr2, CAST, AesFA, AesPA-Net, and our method, together with artifact-sensitive diagnostics for grain and perceptual quality.
- We conduct destructive ablations, step-count sweeps, and $\lambda_{\mathrm{kin}}/\lambda_{\mathrm{swd}}$ scans showing that terminal SWD drives style, kinetic regularization preserves content, and the operating point lies on a smooth trade-off surface rather than a brittle protocol-specific optimum.

Related Work

Classical and arbitrary neural style transfer. Neural style transfer began with optimization-based feature reconstruction and Gram-matrix style losses (Gatys, Ecker, and Bethge 2016). AdaIN later enabled real-time arbitrary transfer by aligning channel-wise feature statistics (Huang and Belongie 2017). Attention-based methods such as SANet improved local style matching through style-attentional feature aggregation (Park and Lee 2019). Recent aesthetic and pattern-aware methods, including AesPA-Net and AesFA, further emphasize pattern repeatability, aesthetic features, and high-quality arbitrary stylization (Hong et al. 2023; Huang et al. 2024). These methods demonstrate the value of feature statistics and local matching, but they generally do not explicitly model a style-conditioned transport path with a terminal distribution constraint.

Transformers and multi-style transfer. Transformer-based style transfer methods such as StyTr2 and S2WAT use attention mechanisms to improve content-style interaction and long-range feature modeling (Deng et al. 2022; Xia et al. 2024). Multi-style approaches amortize many styles in one compact system; SaMST is a recent representative with pluggable style representations, low parameter count, and fast inference (Liu et al. 2024). These systems are directly related to our goal of efficient multi-style transfer. In contrast to representation-only style injection, our method places semantic cross-attention inside a latent bridge objective, so terminal distribution matching and kinetic regularization explicitly shape the style-content trade-off.

Training-free and diffusion methods. Training-free and diffusion-based approaches provide a different route to flexible style transfer. CAST uses VQ autoencoders for content-adaptive training-free style transfer (Gim, Yoo, and Uh 2024), while StyleID injects style into large-scale diffusion models without training (Chung, Hyun, and Heo 2024). These methods can be visually strong, but diffusion sampling and semantic drift can limit their suitability for fast domain-level transfer. Our work targets a complementary efficiency regime: compact latent integration with style-id conditioning rather than iterative diffusion sampling or per-style optimization.

Distribution matching and perceptual evaluation. Our objective is motivated by optimal transport and distribution matching. Sinkhorn distances and computational optimal transport provide efficient tools for approximate transport problems (Cuturi 2013; Peyré and Cuturi 2019), while the Schrodinger problem connects entropic transport to stochastic path measures (Léonard 2014). We use SWD as a lightweight terminal distribution matching objective. For evaluation, we combine perceptual and distributional measures including LPIPS (Zhang et al. 2018), FID (Heusel et al. 2017), KID (Bińkowski et al. 2018), DISTS (Ding et al. 2020), MUSIQ (Ke et al. 2021), MANIQA (Yang et al. 2022), and style-transfer-specific ArtFID (Wright and Ommer 2022). This broad metric set is important because raw CLIP-style and structural scores may fail to penalize grain-like artifacts.

Method

Latent bridge formulation

Let $z_0 \in \mathbb{R}^{4\times 32\times 32}$ denote the VAE latent of a content image and let s denote a target style/domain identifier. The model learns a style-conditioned velocity field

$$v_\theta(z_t, t, s), \tag{1}$$

which transports content latents toward the target artistic domain. At inference time, the endpoint is obtained with explicit Euler integration:

$$z_{t+\Delta t} = z_t + v_\theta(z_t, t, s)\Delta t. \tag{2}$$

The primary evaluation uses 12 integration steps and step size 1.0. We interpret this as a bridge-inspired latent probability flow rather than a strict solver for the continuous-time Schrodinger Bridge problem. The formulation is nevertheless useful: terminal distribution matching controls where the trajectory ends, while path-energy regularization controls how far it moves from the content latent.

Algorithmic view. Training and inference follow the same latent transport backbone but use different supervision paths. During training, a minibatch provides content latents z_0, target style ids s, and target-style latent patches. The network predicts $v_\theta(z_0, 1, s)$, forms the endpoint $\hat{z}_1 = z_0 + v_\theta(z_0, 1, s)$, and optimizes terminal SWD plus kinetic energy. During inference, only the content latent and style id are required; the learned velocity field is integrated for a fixed number of Euler steps and then decoded. This separation is important: training uses endpoint supervision for efficiency, while inference uses multiple small updates to avoid a brittle one-shot transformation.

Architecture

The implementation uses a time-conditioned LBM backbone. The backbone lifts the 4-channel latent to 128 channels, processes high-resolution 32×32 features, downsamples to a 16×16 body representation, applies four semantic cross-attention/body blocks, upsamples, fuses skip features, and decodes a latent velocity. The target style is represented by a learnable spatial prior

$$P_s \in \mathbb{R}^{C\times 16\times 16}, \tag{3}$$

where s ranges over $\{$photo, Hayao, Monet, Van Gogh, Cezanne$\}$. This style prior is a domain-level prototype rather

Table 1: Main architectural and training configuration.

Item	Value
Latent shape	$4 \times 32 \times 32$
Domains	photo, Hayao, Monet, Van Gogh, Cezanne
Base / lift channels	64 / 128
Body resolution	16×16
Body blocks	4 semantic CA blocks
Decoder blocks	2
Style dimension / time dimension	160 / 256
Semantic attention temperature	0.12
Style attention temperature	0.08
Style spatial prior gain	0.35
Terminal SWD weight	20.0
Kinetic weight	1.0
SWD projections	64
SWD patch sizes	3, 5, 7, 15
Primary training length	8 epochs
Primary inference	12 steps, step size 1.0

than a per-reference image representation. It enables fast inference but also limits raw reference-specific style capacity.

Semantic cross-attention injects the style prior into content features:

$$Q = f_c(x), \quad K, V = f_s(P_s), \tag{4}$$

$$A = \mathrm{softmax}(QK^\top/\tau), \quad \hat{x} = AV. \tag{5}$$

The primary model uses semantic attention temperature 0.12, style attention temperature 0.08, and style attention sharpen scale 2.5. Skip fusion uses an additive projected route to preserve spatial content structure.

Training objective

Training uses an OMF-style endpoint approximation. For a content latent z_0 and target style s, the model predicts a velocity at the endpoint training time and forms

$$\hat{z}_1 = z_0 + v_\theta(z_0, 1, s). \tag{6}$$

The loss combines kinetic regularization and terminal SWD:

$$\mathcal{L} = \lambda_{\mathrm{kin}}\mathcal{L}_{\mathrm{kin}} + \lambda_{\mathrm{swd}}\mathcal{L}_{\mathrm{swd}}. \tag{7}$$

The kinetic term is

$$\mathcal{L}_{\mathrm{kin}} = \mathbb{E}\|v_\theta(z_0, 1, s)\|_2^2, \tag{8}$$

which penalizes excessive latent displacement. The terminal SWD term compares generated endpoint patches with target-style patches over 64 sliced projections and patch sizes 3, 5, 7, and 15. The primary configuration uses $\lambda_{\mathrm{kin}} = 1.0$ and $\lambda_{\mathrm{swd}} = 20.0$. Color, cycle, NCE, and repulsive losses are disabled in the main setting. This separation between endpoint style distribution matching and path-energy regularization is the central mechanism behind the observed style-content Pareto behavior.

Why SWD rather than paired reconstruction. The target artistic domain is represented by unpaired examples, so there is no correct pixel-level target for a content image. A paired reconstruction loss would incorrectly force spatial alignment with an unrelated artwork. SWD instead compares projected patch distributions. In one dimension, the Wasserstein distance can be computed by sorting projected samples; averaging over many projections yields an efficient approximation to high-dimensional distribution matching. In our implementation, multi-scale patch sizes make the terminal loss sensitive to both fine texture and broader style statistics.

More concretely, let $\mathcal{P}_r(\hat{z}_1)$ be the set of latent patches of size r extracted from the generated endpoint, and let $\mathcal{P}_r(z_s)$ be the corresponding patch set from target-style samples. For projection directions $\{\theta_m\}_{m=1}^M$, the empirical terminal objective is

$$\mathcal{L}_{\mathrm{swd}} = \sum_{r\in\mathcal{R}} \frac{1}{M} \sum_{m=1}^{M} \big\|\mathrm{sort}\big(\mathcal{P}_r(\hat{z}_1)\theta_m\big) - \mathrm{sort}\big(\mathcal{P}_r(z_s)\theta_m\big)\big\|_2^2. \tag{9}$$

Here $\mathcal{R} = \{3, 5, 7, 15\}$ in the primary configuration. The loss therefore asks whether the generated endpoint has target-style patch statistics, not whether it matches any particular artwork spatially.

Semantic-guided projections. The implementation also supports style-aware projection selection. Rather than relying only on random directions, semantic keys from the style/content routing module identify feature directions with high style response, and SWD is computed along these directions. This gives the terminal loss a closer connection to the style transport module: the same semantic structure used to inject style also influences which endpoint statistics are matched. This is a practical approximation rather than an exact optimal-transport coupling, but it makes the style objective less blind than uniform random projection alone.

Bounded theoretical claim. The method is not an exact Schrodinger Bridge solver. It does not estimate forward and backward stochastic scores, nor does it optimize the full entropic dual objective. The bridge perspective is used to design a tractable training principle: a terminal distribution constraint plus a kinetic path penalty. This bounded interpretation is important for rigor and is also reflected empirically. Removing the terminal constraint weakens style, while removing the path penalty destroys content.

Experiments

Protocol and metrics

We evaluate under a strict 750-output protocol: five source domains times five target domains times 30 source images. The five domains are photo, Hayao, Monet, Van Gogh, and Cezanne; Ukiyo-e is not used in the present protocol. We report CLIP-style, CLIP-content, LPIPS-content, and the composite score

$$\mathrm{EC} = \text{CLIP-style} \cdot (1 - \mathrm{LPIPS}). \tag{10}$$

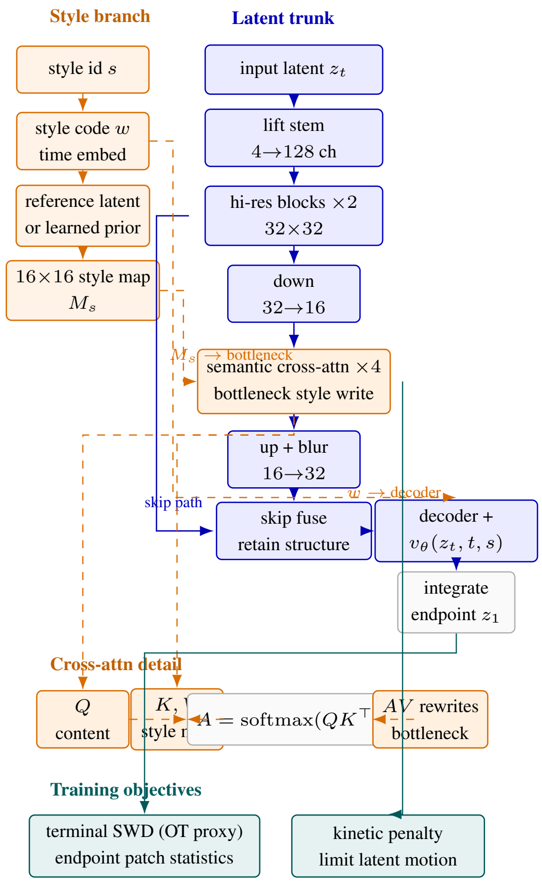

Figure 1: Compact single-column overview of the latent transport model. A U-Net-like latent trunk lifts the input latent, writes style at a 16 × 16 bottleneck through four semantic cross-attention blocks, and predicts a velocity field whose Euler integration yields the styled endpoint z_1. The left branch provides a global style code and a spatial style map; the lower insets expose the core cross-attention mechanism and the two training constraints, namely terminal SWD as an OT-inspired endpoint prior and kinetic regularization as a path-smoothing penalty.

For artifact-sensitive analysis we also report MUSIQ, MANIQA, DISTS-content, high-frequency patch KID, FFT slope error, and shallow Gram micro statistics. We run the strict-750 protocol end-to-end for Ours, SaMST, StyleID, S2WAT, and AdaIN variants.

Metric taxonomy. We use four metric groups. Raw semantic metrics, CLIP-style and CLIP-content, measure whether an output is recognized as stylistically aligned and semantically related to the input. Perceptual content metrics, LPIPS and DISTS-content, measure visual structure preservation. Distributional style metrics, including KID, Gram statistics, and high-frequency patch KID, evaluate whether generated texture statistics resemble target-style images. No-reference image-quality metrics, MUSIQ and MANIQA, help detect dirty or over-processed outputs. We use EC only as an internal Pareto diagnostic for sweep selection; all claims are checked against the individual metrics.

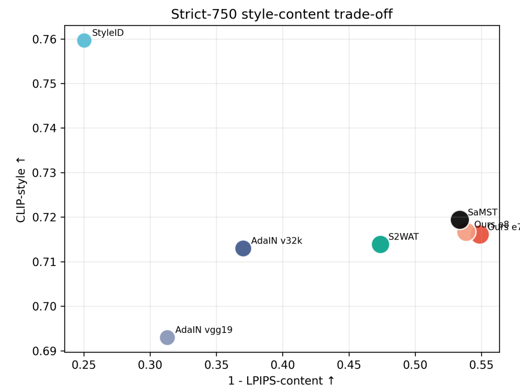

Figure 2: Strict-750 style-content trade-off. Ours is close to SaMST in raw CLIP-style while retaining a better LPIPS-based trade-off than several baselines; StyleID reaches high style but collapses content.

Main comparison

Table 2 shows that Ours epoch 7 remains close to SaMST in raw CLIP-style while improving LPIPS and EC. StyleID obtains the highest CLIP-style, but its CLIP-content of 0.5519 and LPIPS of 0.7497 indicate severe semantic drift. StyTr2, CAST, AesFA, and AesPA-Net all lie clearly below the main frontier under the strict-750 protocol: their CLIP-style remains substantially lower than Ours, SaMST, and StyleID, while their LPIPS ranges from 0.6345 to 0.8215. Overall, LBM occupies a stronger content-preserving trade-off region than these arbitrary-style baselines.

Artifact-sensitive comparison to SaMST

SaMST is a strong baseline under raw style and structure metrics, but the side-by-side grids in Figure 3 reveal muddy and grain-like artifacts. We therefore examine artifact-sensitive diagnostics in Table 3. Ours improves MUSIQ by 13.1109, MANIQA by 0.0918, DISTS-content by 0.0466, HF-Patch-KID by 2.5904, and FFT slope error by 0.5063. These results support the interpretation that SaMST preserves low-frequency structure but produces less realistic local texture statistics.

This diagnostic block addresses a specific evaluation pitfall. Metrics such as CLIP-content, SSIM, and edge overlap tend to reward low-frequency structure preservation. A grainy output can therefore score well if object boundaries remain visible, even when the surface texture is visually dirty. The added diagnostics target a different failure mode: whether local high-frequency patches, no-reference perceptual quality, and frequency slopes resemble real artistic texture rather than spatially incoherent dithering. These metrics should not replace human evaluation, but they provide objective evidence for the artifact mode observed in visual inspection.

Table 2: Strict-750 core comparison. Higher is better for CLIP-S, CLIP-C, and EC. Lower is better for LPIPS. We use EC only as an internal Pareto diagnostic.

Method	CLIP-S	CLIP-C	LPIPS	EC
Ours K1 epoch 7	0.7161	0.8086	0.4514	0.3928
Ours K1 epoch 8	0.7167	0.7977	0.4615	0.3859
SaMST strict	0.7194	0.8193	0.4664	0.3839
S2WAT strict	0.7139	0.7465	0.5263	0.3382
StyleID strict	0.7597	0.5519	0.7497	0.1902
StyTr2 strict	0.6374	0.6001	0.6345	0.2330
CAST strict	0.6652	0.5562	0.7263	0.1821
AesFA strict	0.6496	0.5497	0.7392	0.1694
AesPA-Net strict	0.5976	0.5009	0.8215	0.1067
AdaIN v32k	0.7130	0.6990	0.6298	0.2639
AdaIN vgg19	0.6930	0.5991	0.6870	0.2169

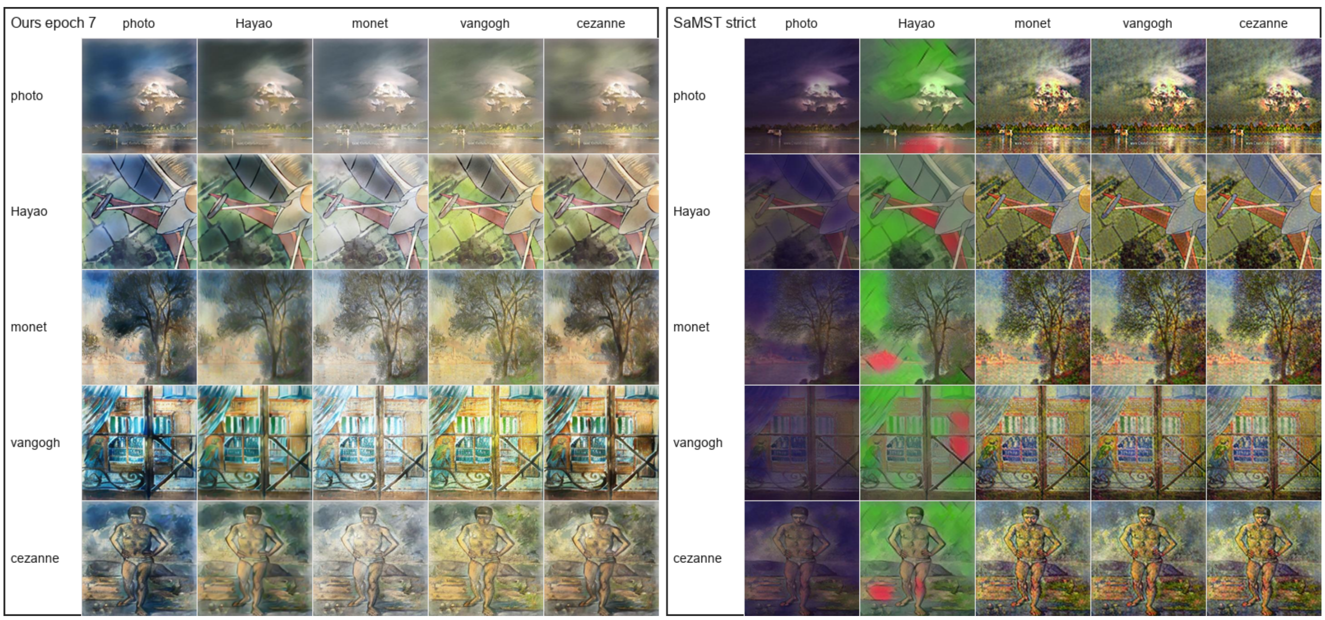

Figure 3: Qualitative 5-by-5 source-target grids for Ours epoch 7 and SaMST strict. Rows indicate source domains and columns indicate target domains. SaMST preserves coarse structure but often produces a muddy, grain-like surface texture; Ours is less aggressive in raw style but visually cleaner and less dithering-like.

Table 3: Artifact-sensitive comparison between Ours and SaMST.

Metric	Ours e7	SaMST
MUSIQ ↑	49.2059	36.0950
MANIQA ↑	0.4057	0.3139
DISTS-content ↓	0.2477	0.2943
HF-Patch-KID ↓	4.1694	6.7598
FFT slope error ↓	0.5473	1.0536
Gram micro ↓	0.0798	0.0947

Destructive ablations

The ablation results in Table 4 expose the mechanism of the model. Removing terminal SWD gives excellent content preservation but weak style, showing that terminal distribution matching is the primary style driver. Removing kinetic regularization preserves style but collapses content, confirming that the path-energy penalty is essential. Strong color loss damages both content and distributional quality, so naive color matching is not used in the mainline.

Table 4: Destructive ablations under the strict-750 protocol.

Variant	CLIP-S	CLIP-C	LPIPS	EC
D0 full	0.7014	0.8022	0.4593	0.3791
D1 w/o terminal SWD	0.6708	0.8989	0.3490	0.4368
D2 w/o kinetic	0.7159	0.6624	0.6375	0.2596
D8 strong color loss	0.6923	0.6629	0.5675	0.2994

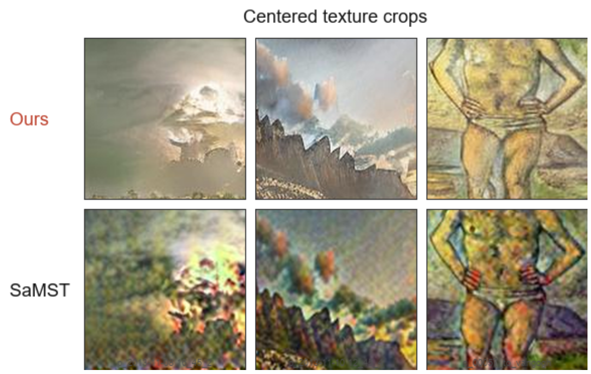

Figure 4: Centered texture crops from matched Ours and SaMST outputs. The crops isolate the surface-texture failure mode: SaMST often retains coarse layout but introduces grain-like, spatially incoherent texture, whereas Ours is visually cleaner.

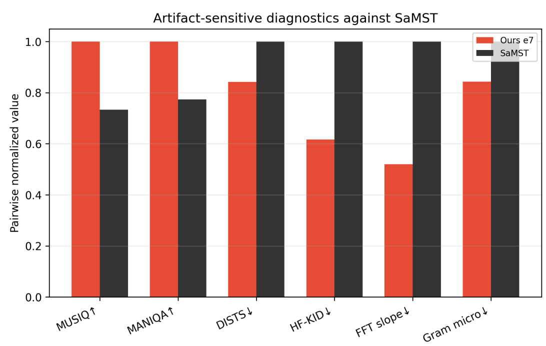

Figure 5: Artifact-sensitive diagnostics for Ours and SaMST. Mixed-scale metrics are pairwise normalized for readability; the raw numeric values are reported in the experimental table.

Weight sweep and theory switches

The 40-run category-weight sweep trained 20 sampling recipes with $K \in \{1, 2\}$ for eight epochs each, evaluating every epoch for 320 total checkpoints. The best EC result was K2 balanced default at epoch 3 with CLIP-style 0.6980, CLIP-content 0.8727, LPIPS 0.3777, and EC 0.4343. The best raw-style result in the sweep was K1 balanced default at epoch 8 with CLIP-style 0.7161, CLIP-content 0.7984, LPIPS 0.4605, and EC 0.3863. The sweep showed that more complex category weighting was not consistently better than balanced sampling.

We also evaluated optional theory switches. Sinkhorn-normalized semantic routing and entropy-gated kinetic regularization improved EC/content but reduced raw CLIP-style. The best theory-switch EC was entropy-gated kinetic with weight 5.0 at epoch 1, giving CLIP-style 0.6916, CLIP-content 0.8804, LPIPS 0.3684, and EC 0.4368. Color transport increased raw style slightly but harmed LPIPS and content, and was therefore not selected as the mainline.

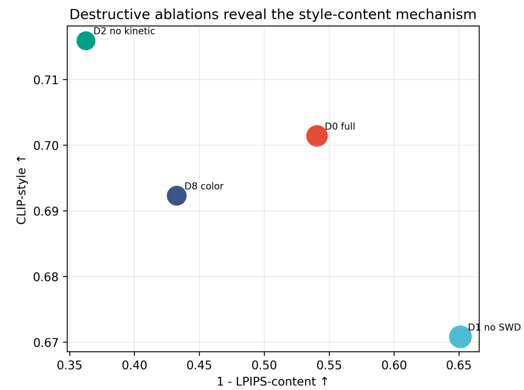

Figure 6: Destructive ablations. Removing terminal SWD improves content but weakens style, while removing kinetic regularization increases style at the cost of severe content degradation.

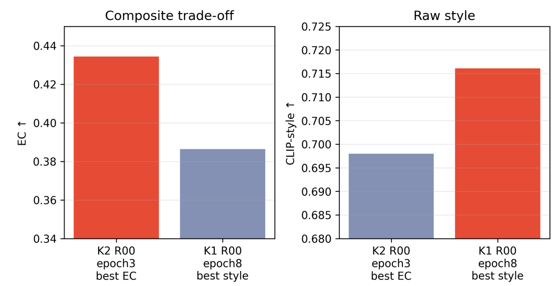

Figure 7: Summary of the 40-run category-weight sweep. K2 balanced default gives the best EC, whereas K1 balanced default gives the best raw CLIP-style among evaluated checkpoints.

Step-count and λ sensitivity

The additional step-count sweep evaluates 1, 4, 8, 12, and 16 Euler steps for the selected epoch-7 checkpoint. The main observation is that the quality curve is almost flat over this range. The all-pairs CLIP-style score varies only from 0.7159 to 0.7162, while LPIPS remains tightly concentrated around 0.4514. The photo-to-art operating point is similarly stable: CLIP-style changes only from 0.6712 to 0.6715 and LPIPS remains near 0.4881. This means the 12-step default is not a brittle tuned optimum. In fact, four steps already recover nearly identical quality, while the measured end-to-end strict-750 throughput increases from 8.43 images/s at one step to 9.54 images/s at four steps and remains around 9.34–9.50 images/s for 8–16 steps. We therefore keep 12 steps as a conservative default, but the sweep indicates that lower-step deployment is viable when marginal latency reduction matters.

We also scanned a 3×3 grid over $\lambda_{\text{kin}} \in \{0.5, 1.0, 2.0\}$ and $\lambda_{\text{swd}} \in \{10, 20, 30\}$ for eight epochs each. The resulting surface is smooth and interpretable rather than erratic. Increasing kinetic weight while reducing terminal pressure systematically improves content preservation: at

Table 5: Unified efficiency profiling. “Prof img/s” measures latent-generator throughput under a common micro-benchmark, whereas “E2E img/s” measures the full strict-750 inference path including file I/O and decoding. Training times marked with † are extrapolated from a single-epoch or single-iteration probe multiplied by the total training length.

Method	Params(M)	FLOPs(G)	Peak VRAM(MB)	Prof img/s	E2E img/s	Train sec
Ours (12-step latent gen.)	3.91	–	33.4	102.16	9.34	309.9
SaMST	–	–	–	–	–	6768.7†
S2WAT	–	–	–	–	–	10600†
StyleID	training-free	–	–	–	–	0

epoch 8, the setting $(2.0, 10)$ yields CLIP-content 0.8734, LPIPS 0.3720, and EC 0.4373, compared with the default $(1.0, 20)$ setting at CLIP-content 0.8124, LPIPS 0.4477, and EC 0.3937. Conversely, weaker kinetic weight or stronger terminal weight pushes the model toward more aggressive but less content-faithful stylization; for example, $(0.5, 30)$ drops CLIP-content to 0.6926 and raises LPIPS to 0.5538. The best transfer-EC checkpoints in this grid occur consistently around epoch 3, again showing that the protocol exposes a broad style-content frontier rather than a single fragile endpoint.

Resource use and reproducibility

For efficiency, we distinguish two measurement scopes. The first is an end-to-end strict-750 measurement that includes generation, decoding, file I/O, and result collation. The second is a unified forward micro-benchmark that profiles the stylization network itself under a fixed hardware/software stack. This separation matters because file-system overhead can partially mask step-count differences in end-to-end runs.

All reported strict-750 evaluations use the same domain protocol and preprocessing pipeline. We also evaluated every epoch in the 40-run sweep, yielding 320 checkpoint evaluations. Final-epoch selection alone can hide style-content trade-offs: K2 balanced default achieved the best EC at epoch 3, whereas K1 balanced default achieved the best raw style at epoch 8.

Discussion and Limitations

The experiments support a specific and conservative conclusion. The proposed method does not universally dominate all baselines on raw style: StyleID has higher CLIP-style, and SaMST is slightly higher than Ours epoch 7 on raw CLIP-style and CLIP-content. However, StyleID suffers content collapse, while SaMST exhibits grain-like artifacts that are poorly captured by standard structure metrics. StyTr2, CAST, AesFA, and AesPA-Net also fall clearly below the main trade-off frontier. Ours therefore provides a cleaner and more balanced operating point rather than an across-the-board absolute maximum on every scalar metric.

The current study still has several limitations. First, the style representation is domain-level rather than reference-image-specific; this makes inference efficient but limits the ability to reproduce unique single-image brushstroke patterns or resolve cross-domain prototype conflicts. Second, Euler integration is deliberately simple, so discretization error can accumulate when the latent path is extrapolated to more aggressive settings or higher resolutions. Third, the SWD term uses heuristic multi-scale patch projections rather than an exact transport solver, so the bridge interpretation should remain bounded and practical rather than fully theoretical. Fourth, although the core metrics and efficiency profiling are complete for the compared baselines, the full artifact-diagnostic pack has only been rerun for a subset of methods, and user-study data has not yet been collected. Finally, high-resolution generalization beyond the present 256-based protocol remains under-validated. Future work should evaluate richer domain prototype priors, controlled integration horizons, higher-resolution extrapolation, and human preference studies.

Conclusion

We presented a latent bridge-inspired framework for efficient multi-style artistic transfer. By combining terminal SWD style distribution matching with kinetic path regularization, the method separates the forces that drive style from those that preserve content. A lightweight style spatial prior and semantic cross-attention make this formulation practical for efficient domain-level transfer. Under a strict 750-output evaluation, the method reaches a competitive raw style score while improving LPIPS and artifact-sensitive perceptual diagnostics over SaMST and preserving content substantially better than StyleID. The results indicate that latent bridge matching is a promising foundation for clean, efficient, and reproducible artistic style transfer.

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Reproducibility Checklist

This paper

- Includes a conceptual outline and/or pseudocode description of AI methods introduced: **Yes**.
- Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results: **Yes**.
- Provides well marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper: **Yes**.

Does this paper make theoretical contributions? No. The paper uses a bounded bridge-inspired motivation rather than theorem-level theoretical claims.

Does this paper rely on one or more datasets? Yes.

- A motivation is given for why the experiments are conducted on the selected datasets: **Yes**.
- All novel datasets introduced in this paper are included in a data appendix: **NA**.
- All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **NA**.
- All datasets drawn from the existing literature are accompanied by appropriate citations: **Yes**.
- All datasets drawn from the existing literature are publicly available: **Yes**.
- All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing: **NA**.

Does this paper include computational experiments? Yes.

- This paper states the number and range of values tried per (hyper-)parameter during development of the paper, along with the criterion used for selecting the final parameter setting: **Partial**.
- Any code required for pre-processing data is included in the appendix: **No**.
- All source code required for conducting and analyzing the experiments is included in a code appendix: **No**.
- All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes: **Partial**.
- All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from: **Partial**.
- If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results: **Partial**.
- This paper specifies the computing infrastructure used for running experiments, including hardware/software details: **Partial**.
- This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics: **Yes**.
- This paper states the number of algorithm runs used to compute each reported result: **Yes**.
- Analysis of experiments goes beyond single-dimensional summaries of performance to include measures of variation, confidence, or other distributional information: **Yes**.
- The significance of any improvement or decrease in performance is judged using appropriate statistical tests: **No**.
- This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments: **Partial**.